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0178/01

October/November 2017

1 hour

Additional Materials: Geometrical Instruments
Tracing Paper (optional)

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO NOT WRITE IN ANY BARCODES.

Answer **all** questions.
If working is needed for any question it must be shown below that question.

ELECTRONIC CALCULATORS **MUST NOT** BE USED IN THIS PAPER.

The number of marks is given in brackets [] at the end of each question or part question.
The total marks for this paper is 60.

This document consists of **11** printed pages and **1** blank page.



1 Evaluate

(a) $\frac{2}{3}$ of 69,

Answer (a) [1]

(b) $2\frac{1}{3} - 1\frac{2}{7}$,

Answer (b) [2]

(c) $4\frac{1}{2} + 27\frac{1}{3} + 5^\circ$.

Answer (c) [2]

2 Insert one pair of brackets in the following statement to make it true.

$$3 + 2 \times 4 \div 5 - 1 = 5$$

Answer [1]

3 Write 298 correct to 2 significant figures.

Answer [1]

- 4 Given that $f(x) = 1 - 3x$, find $f(2)$.

Answer [2]

- 5 (a) Express 56 as the product of its prime factors.

Answer (a) [1]

- (b) The number P expressed as the product of its prime factors is $2 \times 3 \times 7$.
Find the HCF of 56 and P .

Answer (b) [1]

- 6 $w = 3 \times 10^3$ and $z = 5 \times 10^2$.

Work out the following, giving your answers in standard form.

- (a) wz

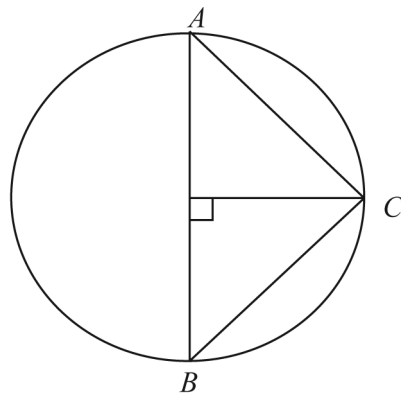
Answer (a) [2]

- (b) $w - z$

Answer (b) [1]

- 7 Points A , B and C lie on the circumference of a circle, centre O .

$\angle BOC = 90^\circ$ and $AB = 10$ cm.



NOT TO
SCALE

- (a) State with a reason the value of angle OBC .

Angle $OBC =$ because
..... [2]

- (b) Calculate AC , giving your answer correct to the nearest integer.

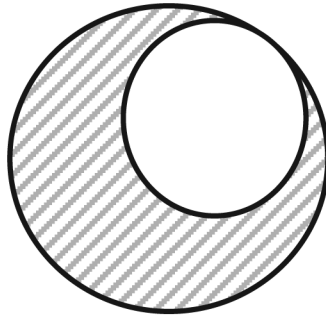
Answer (b) cm [2]

(c) State the **special** name of the triangle ABC .

Answer (c) [1]

8 The diagram shows two circles.

The radius of the smaller circle is 3 cm and that of the larger circle is 5 cm.



NOT TO
SCALE

Calculate, in terms of π

(a) the area of the smaller circle,

Answer (a)cm² [1]

(b) the area of the shaded region.

Answer (b)cm² [2]

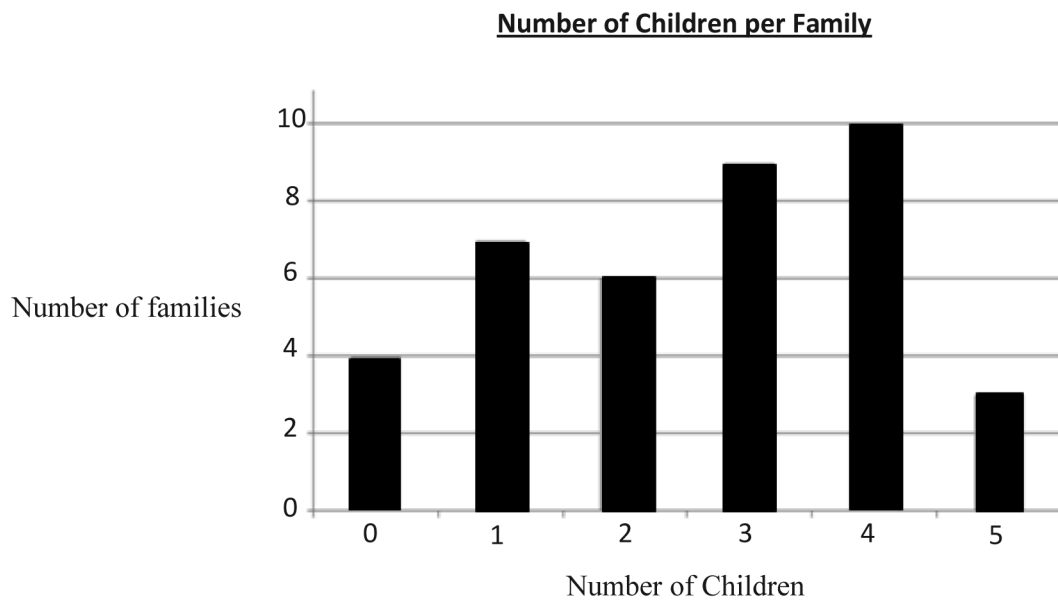
- 9 Solve the simultaneous equations.

$$\begin{aligned} p - 2q &= -1 \\ 3p + q &= 11 \end{aligned}$$

Answer $p =$

$q =$ [3]

- 10 The bar chart shows the number of children per family in a small village.



- (a) Calculate the total number of families in the village

Answer (a) [2]

- (b) Find the median number of children per family

Answer (b) [2]

- (c) A family is chosen at random
Find the probability that it has more than 3 children.

Give your answer in its simplest form.

Answer (c)% [2]

- 11 Express 12 cm as a percentage of 4 m.

Answer% [2]

- 12 Write the ratio 5.5 : 2.5 in its simplest form.

Answer [1]

13 $A = \begin{pmatrix} 2 & 0 \\ 3 & 1 \end{pmatrix}, B = \begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix}$

(a) Find $A - 2B$.

Answer (a) [2]

(b) Given that $A \begin{pmatrix} x \\ 2 \end{pmatrix} = \begin{pmatrix} 8 \\ -2y \end{pmatrix}$, find the values of x and y .

Answer (b) $x = \dots\dots\dots y = \dots\dots\dots$ [3]

14 (a) Simplify $3x(2x + 3) - (9x + 5)$.

Answer (a) [2]

(b) Solve the equation.

$$3a - 12 = a - 2$$

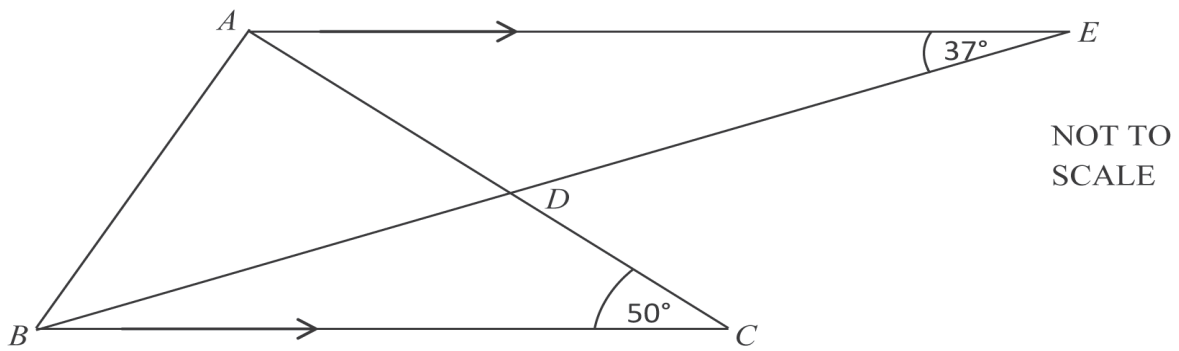
Answer (b) $a = \dots\dots\dots$ [2]

(c) Express as a single fraction in its simplest form.

$$\frac{2c-1}{3} - \frac{c-2}{2}$$

Answer (c) [3]

- 15 The diagram shows figure $ABCDE$.
 $AC = BC$, $\angle AEB = 37^\circ$, $\angle ACB = 50^\circ$ and AE is parallel to BC .



- (a) Find angle CAE and give the reason for your answer.

Angle CAE because
 [2]

- (b) Find angle ADE and give the reason for your answer.

Angle ADE because
 [2]

16 $\mathcal{C} = \{2, 3, 4, 5, 6, 7, 8, 9, 10\}$

$$A = \{x : x \text{ is an even number}\}$$

$$B = \{x : x \text{ is a factor of } 10\}.$$

(a) List the elements of set

(i) A ,

Answer (a)(i) [1]

(ii) $A' \cap B$.

Answer (a)(ii) [1]

(b) Find $n(A \cup B)$.

Answer (b) [2]

17 The columns, n , P , Q and S show 4 sequences

n	P	Q	S
1	10	5	15
2	16	8	24
3	22	11	33
4	----	----	----

(a) Fill in the 3 missing numbers in the 4th row. [3]

(b) Write a formula connecting

(i) P , Q and S ,

Answer (a)(i) [1]

(ii) n and Q .

Answer (a)(ii) [2]

